HEALTHIER, LONGER, BETTER LIVES

# Medical Receipt OCR Benchmark

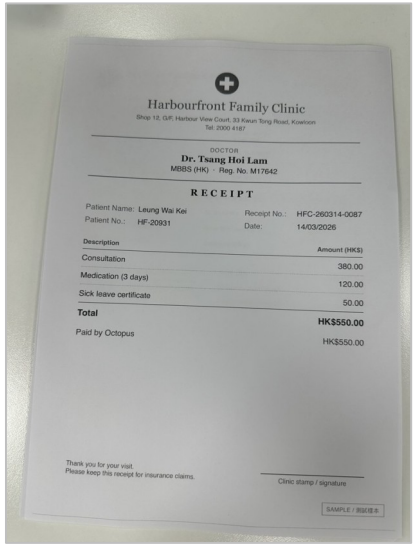
Can Azure OpenAI replace hand-written extraction rules?

9 receipts · 6 fields · 4 approaches

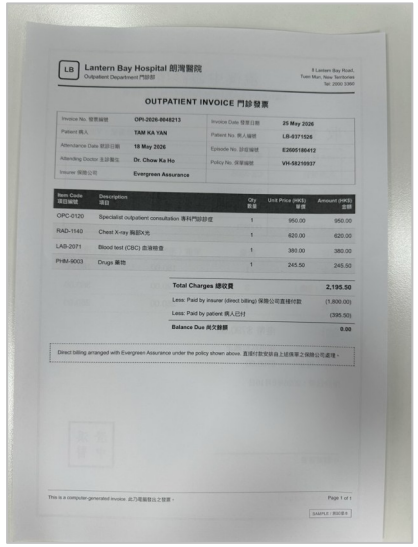
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# What We Tested

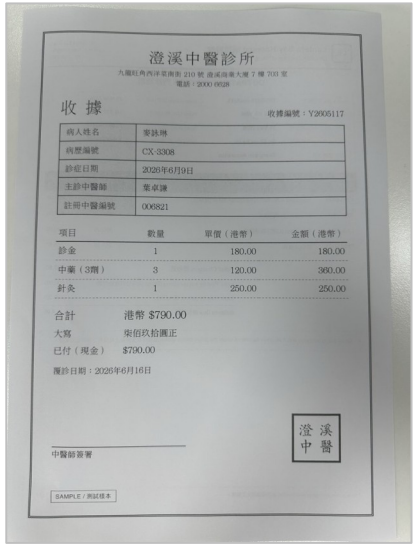
9 receipts with different layouts and languages. Each approach reads the text, then finds 6 fields.




English




Bilingual




Chinese

| Approach | 1. Read text | 2. Find fields |
|---|---|---|
| **PaddleOCR Light** | Open-source OCR | Rules |
| **PaddleOCR Heavy** | Open-source OCR | Rules |
| **Azure DI** | Azure OCR | Rules |
| **Azure DI + AOAI** | Azure OCR | Azure OpenAI |

Fields: doctor, registration no., patient, receipt no., service date, total.
51 fields scored.


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# Results: Same Accuracy, Different Effort

Rules reached 88.2% only after manual tuning. DI + AOAI reached 88.2% with its prompt unchanged.

| Approach | Original rules | Improved rules* | Time per receipt |
|---|---|---|---|
| **PaddleOCR Light** | 37.3% (19/51) | **88.2%** (45/51) | 13.9 s |
| **PaddleOCR Heavy** | 29.4% (15/51) | 64.7% (33/51) | 53.3 s |
| **Azure DI** | 41.2% (21/51) | **86.3%** (44/51) | 3.8 s |
| **Azure DI + AOAI** | **88.2%** (45/51) | No rules (prompt unchanged) | 6.2 s |

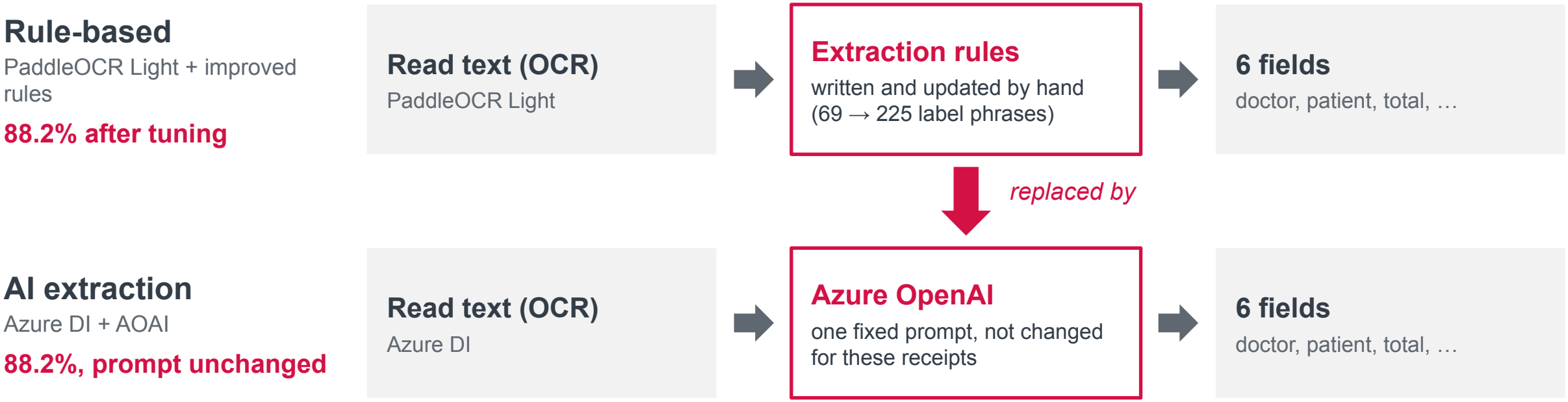
** Tuned using these 9 receipts, so not a measure of accuracy on new receipts. Accuracy = correct fields out of 51. All approaches processed all 9 receipts.*


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# AOAI Replaces the Manual Rule Layer

Both pipelines read the text the same way. Only the step that finds the fields changes.



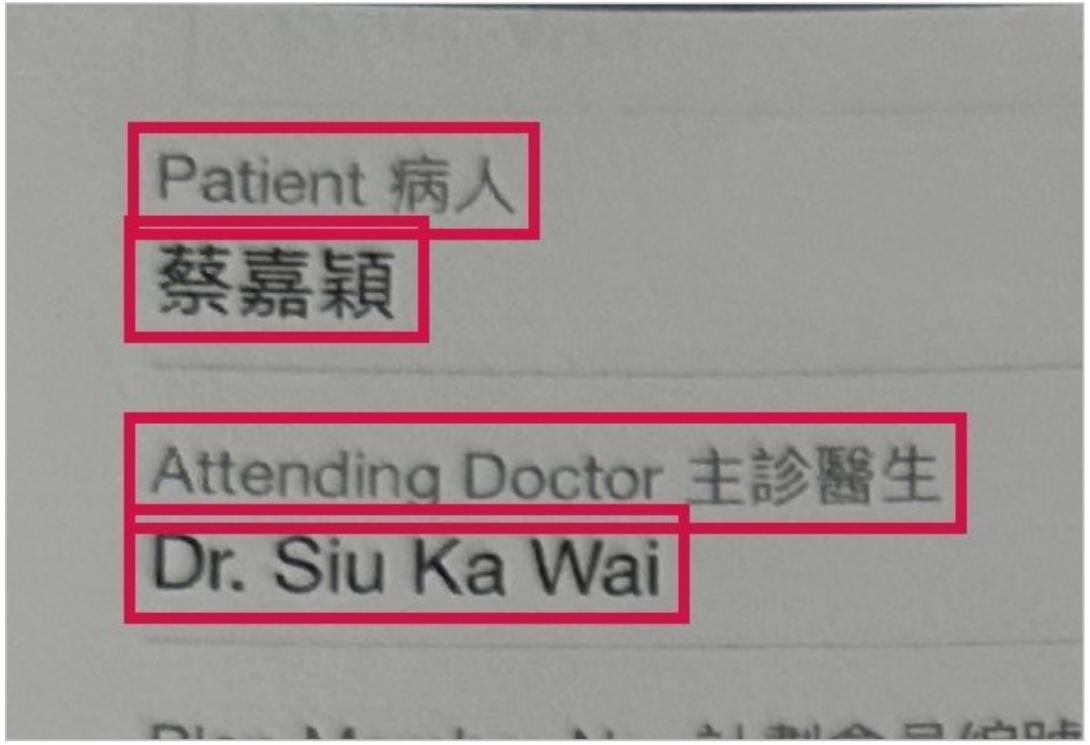

**AOAI replaces the manual rule layer.** It still makes mistakes (6 of 51 fields wrong) and has not yet been tested on a large set of unseen receipts.


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# Example 1: The OCR Was Right, the Rules Were Wrong

Both OCR engines read the names. The original rules took the label 病人 as the patient's name.




| Field | Correct | Original rules | Improved rules | DI + AOAI |
|---|---|---|---|---|
| **Patient** | 蔡嘉穎 | ✗ 病人 | ✓ 蔡嘉穎 | ✓ 蔡嘉穎 |
| **Doctor** | Dr. Siu Ka Wai | ✗ not found | ✓ Dr. Siu Ka Wai | ✓ Dr. Siu Ka Wai |

- Original rules: took the Chinese label 病人 ("patient") as the patient's name.
- Improved rules: fixed by adding bilingual labels by hand.
- AOAI: correct with the same prompt; no change needed.

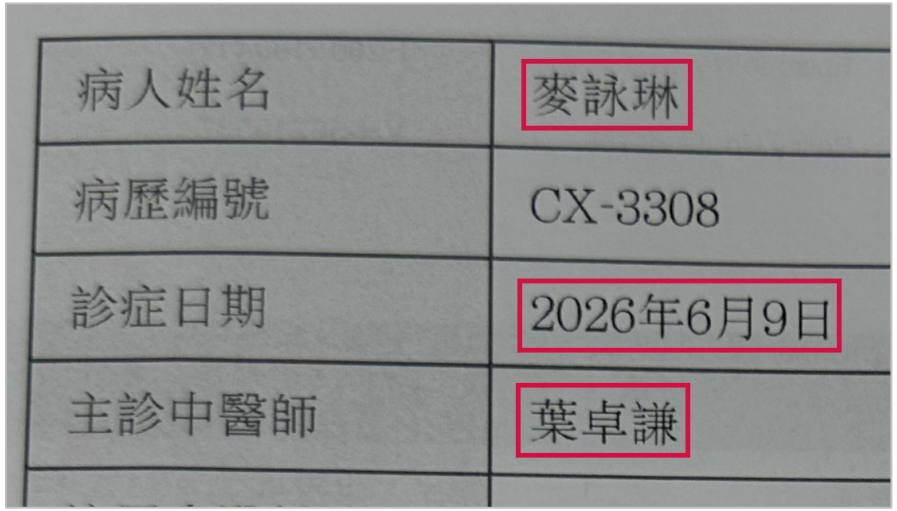
*Bilingual medical centre receipt. Rules shown on PaddleOCR Light text; Azure DI with the original rules made the same patient error.*


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# Example 2: Reaching 88.2% With Rules Took Manual Work

Each new language or label wording had to be added by hand, and one change broke another receipt.




Chinese medicine receipt: original rules 0/6 fields, improved rules 6/6, DI + AOAI 6/6 (prompt unchanged).

**69 → 225**

label phrases in the rules, added by hand after reviewing the errors

**369303 → 380048**

vet invoice receipt number: correct before the changes, wrong after (a regression)

**New layouts**

will likely need more rule changes and a retest of every receipt

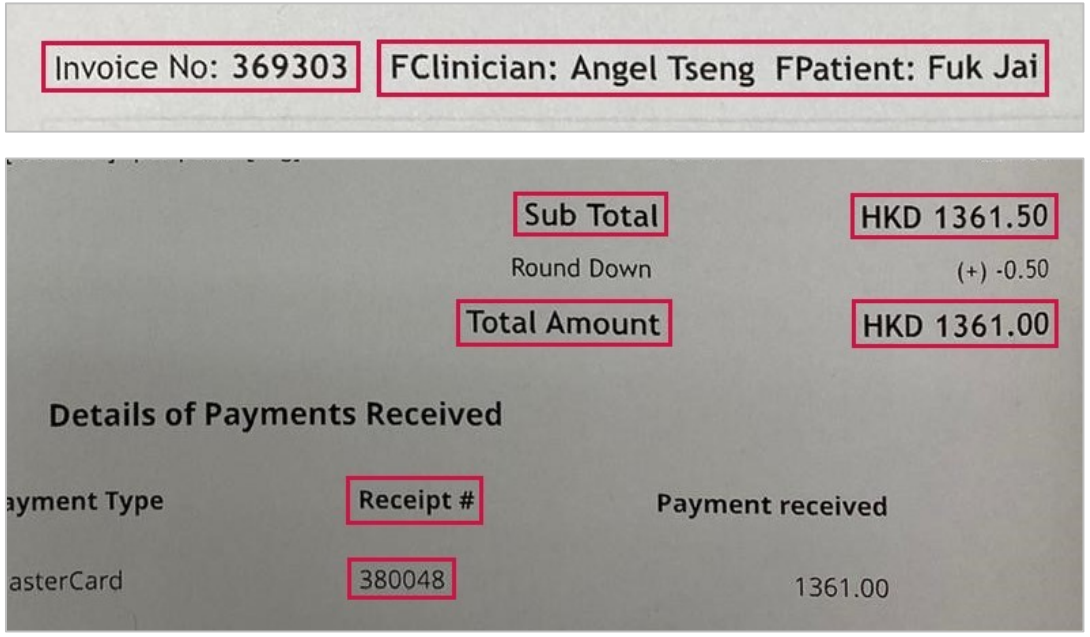
DI + AOAI needed none of these changes for these receipts. Whether that holds for new layouts still needs to be tested.


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# Example 3: AOAI Still Makes Mistakes

On the vet invoice, the improved rules and AOAI each got a different field wrong.




| Field | Correct | Original rules | Improved rules | DI + AOAI |
|---|---|---|---|---|
| **Receipt no.** | 369303 | ✓ 369303 | ✗ 380048 | ✓ 369303 |
| **Patient** | Fuk Jai | ✗ not found | ✗ not found | ✓ Fuk Jai |
| **Total** | 1361 | ✓ 1361.0 | ✓ 1361.0 | ✗ 1361.5 |

- Improved rules: picked the payment receipt number.
- Rules: the OCR read "FPatient:", so the patient label was missed.
- AOAI: picked the sub total instead of the total.


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# What It Takes to Reach 88.2%

Same accuracy. The difference is the manual work to get there, and to keep it there.

| | PaddleOCR Light + improved rules | Azure DI + AOAI |
|---|---|---|
| **Accuracy (9 receipts)** | **88.2% (45/51)** | **88.2% (45/51)** |
| **Effort to get there** | Rules rewritten by hand after reviewing errors on these receipts | None: prompt unchanged |
| **Extraction logic** | 69 → 225 label phrases to maintain | One fixed prompt |
| **New receipt layouts** | Likely more rule updates and a regression test of every receipt | Possibly no change; not yet tested |
| **Errors seen** | 6 wrong fields, incl. one regression caused by tuning | 6 wrong fields, e.g. the vet invoice total |
| **Time and running cost** | 13.9 s, own compute, no API fee | 6.2 s, Azure fees per receipt (to confirm) |

*The main question is not API price but who maintains the extraction logic. Lower maintenance for AOAI is implied by this benchmark, not yet proven.*


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# Pure OCR + Rules vs. Azure DI + AOAI: Pros and Cons

Both reached 88.2%, AOAI with no prompt changes. "Measured" = this benchmark; the rest is expected.

## Pure OCR + Rules (PaddleOCR Light)

**New formats may need manual rule updates, which can break working cases and add maintenance and testing.**

### Pros

- **Measured:** 88.2% after manual rule tuning (from 37.3%).
- Full control; predictable on formats the rules know.
- No API fees: open-source OCR on our own compute.

### Cons

- New formats and layouts may need rule updates by hand.
- **Measured:** a rule change broke a receipt that was already correct.
- **Measured:** rules grew from 69 to 225 label phrases.

## Azure DI + AOAI

**Less rule maintenance expected: accuracy depends mainly on the AI model, OCR output and prompt.**

### Pros

- **Measured:** 88.2% with no prompt changes.
- Reads varied formats using existing model capabilities.
- **Measured:** faster, 6.2 s vs 13.9 s per receipt.

### Cons

- **Measured:** still wrong on 6 of 51 fields, e.g. a total.
- Less direct control: results depend on model, OCR, prompt.
- API fees; needs validation, monitoring and prompt upkeep.

**Both reach 88.2% here.** The difference is where the maintenance effort sits, and how much of it we own.


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# Conclusion and Next Steps

The real trade-off: maintain extraction rules ourselves, or let AI replace that layer.

- PaddleOCR Light + improved rules and Azure DI + AOAI both reached 88.2% on this benchmark.
- PaddleOCR Light needed manual rule tuning to get there. Azure DI + AOAI reached the same accuracy without changing its prompt.
- **So the choice is whether to keep writing and maintaining extraction rules, or to use AI for that layer.**

### 1. Freeze both

Lock the improved rules and the AOAI prompt.

### 2. Test on unseen receipts

30+ real, anonymised receipts. Measure accuracy and count the rule changes needed.

### 3. Then compare cost

Compute, Document Intelligence pages and gpt-5-mini tokens.

*Limitation: 9 receipts, mostly printed mock receipts; the improved rules were tuned on them. AOAI still makes mistakes and has not been tested on a large unseen set.*


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